

Sparse gestures

Each phone has G gesture programs: G = 6 or 8.

$$\theta_{pg} = (y_{pg}, \omega_{pg}, t_{pg}^{\text{on}}, t_{pg}^{\text{off}})$$

target · speed · start · end

1. Eight-row gesture vector

AA in “more” · L2 · same start and +3,000 training steps

Gesture program		$\lambda = 0.02$	$\lambda = 0.05$
$\theta_{p,1}$	Gesture 1	1	0
$\theta_{p,2}$	Gesture 2	1	0
$\theta_{p,3}$	Gesture 3	1	1
$\theta_{p,4}$	Gesture 4	1	0
$\theta_{p,5}$	Gesture 5	1	0
$\theta_{p,6}$	Gesture 6	1	1
$\theta_{p,7}$	Gesture 7	1	1
$\theta_{p,8}$	Gesture 8	1	1

1 = ON; 0 = OFF. M and R are OFF in both saved runs.

Avg gestures in “more”:

$$\frac{0+8+0}{3} = 2.67 \longrightarrow \frac{0+4+0}{3} = 1.33$$

2. Sparsity loss — λ is the penalty weight

$$\mathcal{L} = \mathcal{L}_{\text{EMA}} + 2\mathcal{L}_{\text{velocity}} + 0.5\mathcal{L}_{\text{TV}} + \lambda\mathcal{L}_{\text{gate}}$$

$$\mathcal{L}_{\text{gate}} = \frac{\sum_{p,g} pm_p q_{pg}}{\sum_p pm_p + 10^{-6}}, \quad q_{pg} = \sigma(\ell_{pg})$$

$$D = \sum m + 1, \quad \mathcal{L}_{\text{EMA}} = \frac{\sum m \text{mean}_{12}[(\hat{E} - E)^2]}{D}, \quad \mathcal{L}_{\text{velocity}} = \frac{\sum_{t \geq 1} m_t \text{mean}_{12}[(\Delta \hat{E} - \Delta E)^2]}{D}$$

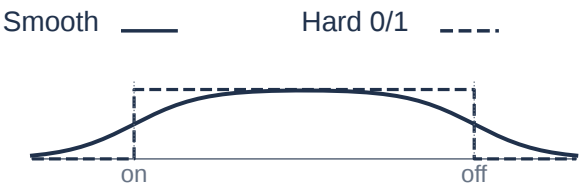
$$\mathcal{L}_{\text{TV}}^{\text{L1/line, undef}} = \frac{\sum mA(X - Z)^2}{\sum mA + 1}, \quad \mathcal{L}_{\text{TV}}^{\text{L3/all}} = \frac{\sum m \text{mean}_G[(\hat{Z} - Z)^2]}{D}, \quad \mathcal{L}_{\text{TV}}^{\text{L2}} = 0$$

m: valid loss frames; pm: valid phones; E, Z: proxy references (z units).

L1 probability penalty ≠ L1 model. No /G or window weighting.

$$h_{pg} = \mathbf{1}[\ell_{pg} > 0] \mathbf{1}[\max_t a_{pg}(t) \geq 0.5], \quad \text{Avg gestures} = \frac{\sum_{p,g} pm_p h_{pg}}{\sum_p pm_p}$$

3. Window shapes



Schematic · hard-window test pending

Fewer gestures with matched quality: indirect evidence; random-control and interpretation checks pending.

Matched results: pending

val256 · L2, G = 8

λ	Avg / phone	WER	PER
0.02	Pending	Pending	Pending
0.05	Pending	Pending	Pending